

اختبار ميد عام 2010، لمادة قواعد البيانات

CPIT240

الملف جابه لنا طالب متخرج

إن شاء الله يفيدكم

FCIT20:

<https://t.me/FCIT20Male>

Mid Term Exam

Course Title: Database Systems – CPIT240
Program: BS (IT) :
Resource Person:
min

Total Marks: 80 (20+60)
Mid Term (Nov. 2010)
Total Time allowed: 120

Name:

Roll No:

PART I (OBJECTIVE)

Instructions:

- Write your Name, Student Id and Section on subjective question.
- Cutting and overwriting is not allowed otherwise negative marks will be calculated for such objective answers.
- Do not write anything on Objective Question papers except your Name, Roll Number & Section.
- Both Objective and Subjective question papers and objective answer sheet have to be returned to the invigilation staff before leaving the examination hall.

Question (1)

Marks (20)

Encircle True or False on answer sheet provided to you for objective questions 1 to 10:

1. A database application is a system program that is used to perform a series of activities on behalf of database users. (True)
2. Enterprise data model does not describe the high-level entities in an organization and the relationship between these entities. (False)
3. A database is a self-describing collection of integrated records. (True)
4. Data that describes the properties and context of user data is known as Metadata. (True)
5. Entity type that is not existence-dependent on some other entity type is called a weak entity. (False)

6. The most common degree for relationships is binary. (True)
7. Where a model represents a relationship between entity types, but pathway between certain entity occurrences is ambiguous is called chasm trap. (True)
8. A user of a database system can be an entity. (False)
9. A relationship which also has attributes is called an associative entity. (True)
10. In DBMS architecture the conceptual level does not present a logical view of the entire database. (False)

PART I (OBJECTIVE)

For following question 11 to 20 encircle only one correct answer on the answer sheet provided to you.

11. The hierarchical data model organizes data in a _____ structure.
 a) Tree b) Object Relational c) Object Oriented d) Relational
12. The first stage in the design of a database is to define the _____ view.
 a) External b) Conceptual c) Internal d) Physical
13. A software system that enables users to define, create, maintain, and control access to the database is called _____.
 a) File System b) Operating System c) DBMS d) Application Program
14. Primary key attribute should contain _____ values.
 a) Duplicate b) Unique c) Single d) Null
15. Associative entity is a combination of relationship and _____.
 a) Attributes b) Cardinality c) Foreign Key d) Candidate Key
16. Degree of a relationship is represented by the number of _____.
 a) Relationship b) Users c) Entities d) Keys
17. Recursive relationship is of degree _____.

- a) Quaternary b) Binary c) Ternary d) Unary

18. Entity type that is existence – dependent on some other entity type is called _____.

- a) Strong Entity b) Weak Entity c) Parent Entity d) Zero Entity

19. A row in a relation is also called _____.

- a) Record b) Tuple c) Instance d) All of the above

20. Oracle is a _____.

- a) DBMS b) Database c) Tuple d) Relation

PART 2 (SUBJECTIVE)

Marks(60)

Instructions:

- For following Questions 2 to 7 write your answers in the space provided to you on question paper.
- Question 7 – Optional.
- Both Objective and Subjective question paper sheet have to be returned to the invigilation staff before leaving the examination hall.

Question (2)

Marks(10)

Differentiate between the followings:

1. Two tier and Three tier architecture

Two tier: The two-tier architecture is like client server application. The direct communication takes place between client and server. There is no intermediate between client and server.

Three tier : Three tier architecture having three layers. They are

1. Client layer
2. Business layer

3. Data layer

Client layer: Here we design the form using textbox, label etc.

Business layer: It is the intermediate layer which has the functions for client layer and it is used to make communication faster between client and data layer. It provides the business processes logic and the data access.

Data layer: it has the database.

2. Attribute and Tuple

TUPLE:

Is the "ROW" in a table and

ATTRIBUTE:

Is the "COLUMN" and it can also be called as "ATTRIBUTE".

3. Stored (Simple) Attribute versus Derived Attributes

Stored Attribute: An attribute that supplies a value to the related attribute.

Derived Attribute: An attribute that's value is derived from a stored attribute.

4. Data and Information

Data: is raw material & unorganized facts that need to be processed

Information: When data are processed, organized, structured or presented in a given context so as to make them useful, they are called Information.

5. Metadata and database

Metadata: that describes the fields and formats of databases and data

Database: is a collection of information that is organized so that it can easily be accessed, managed, and updated. In one view, databases can be classified according to types of content: bibliographic, full-text, numeric, and images.

Question (3)**Marks (14)**

(i) Make the relationships between the given tables (relations) on the following diagram based on the referential integrity constraints for company database schema and identify the primary and foreign keys on the diagram also.

(ii) Also write the referential integrity constraint (Equations) for the following company database schema.

Employee

FNAME	MINIT	LNAME	<u>SSN</u>	BDATE	ADDRESS	SEX	SALARY	SUPER_SSN	DNO
-------	-------	-------	------------	-------	---------	-----	--------	-----------	-----

DEPARTMENT

DNAME	<u>DNUMBER</u>	MGR_SSN	MGR_START_DATE
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DEPT_LOCATIONA

<u>DNUMBER</u>	<u>DLOCATION</u>
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PROJECT

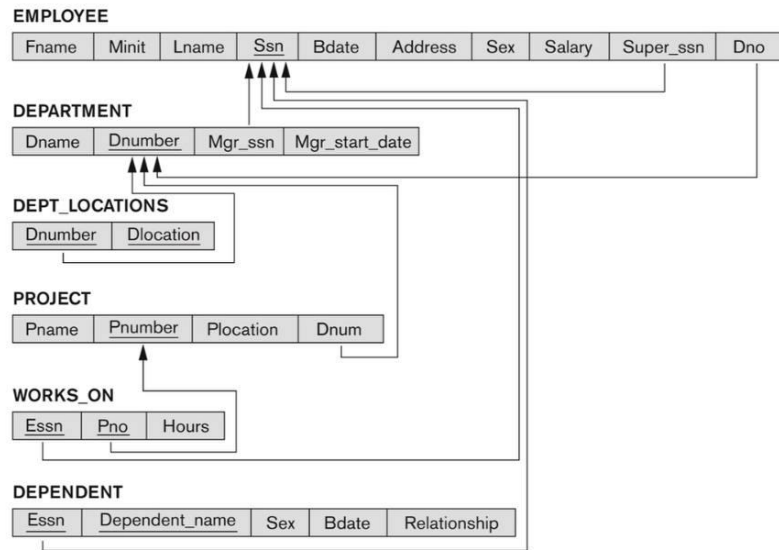
PNAME	<u>PNUMBER</u>	PLOCATION	DNUM
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WORKS ON

<u>ESSN</u>	<u>PNO</u>	HOURS
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DEPENDENT

<u>ESSN</u>	<u>DEPENDENT_NAME</u>	SEX	BDATE	RELATIONSHIP
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Equations:

FK(employee.super_ssn) = PK (employee.ssn)
FK(Department.Mgr_ssn) = PK (employee.ssn)
FK(employee.dno) = PK (Department.Dnumber)
FK(Dept_Location.Dnumber) = PK (Department.Dnumber)
FK(Works_On.Essn) = PK (employee.ssn)
FK(Dependent.Essn) = PK (employee.ssn)
FK(Project.Dnum) = PK (Department.Dnumber)
FK(Works_On.Pno) = PK (Project.Pnumber)

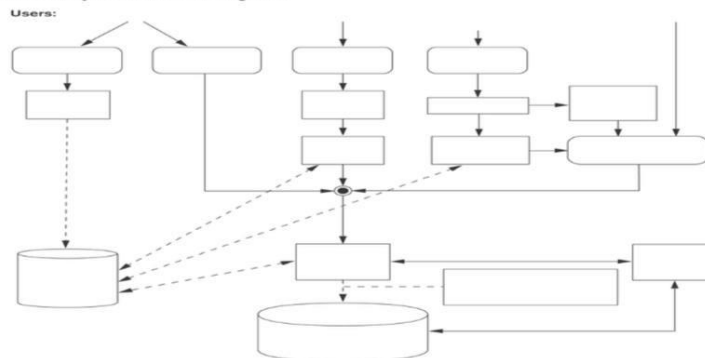
Question (4)**Marks (8)**

Based on the constraints that have been studied, Arrange the fields below by selecting any of the following should be entered first, and then the next, and so on?

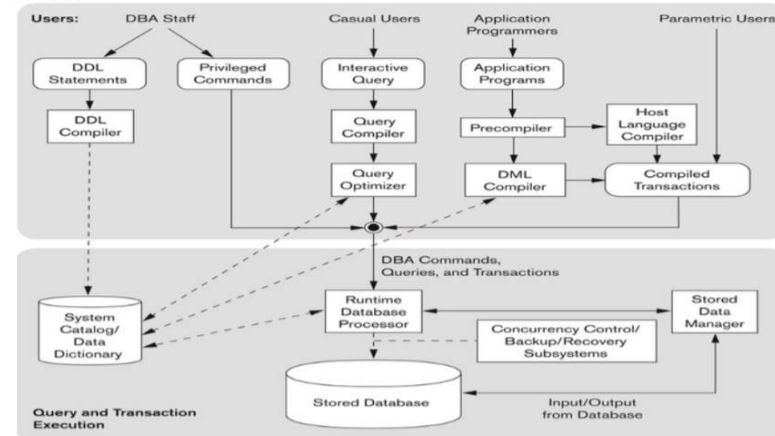
EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
5- 7369	SMITH	CLERK	7788	17-DEC-80	800		20
6- 7499	ALLEN	SALESMAN	7369	20-FEB-81	1600	300	30
7- 7521	WARD	SALESMAN	7499	22-FEB-81	1250	500	30
2- 7566	JONES	MANAGER	7782	02-APR-81	2975		20
4-8- 7654	MARTIN	SALESMAN	7698	28-SEP-81	1250	1400	30
3- 7698	BLAKE	MANAGER	7566	01-MAY-81	2850		30
1- 7782	KING	PRESIDENT		09-JUN-81	2450		10
4- 7788	SCOTT	ANALYST	7698	19-APR-87	3000		20

Question (5)**Marks (10)**

Fill a diagram for the Component modules of a DBMS and properly write the names of the components on this diagram.



Solution:



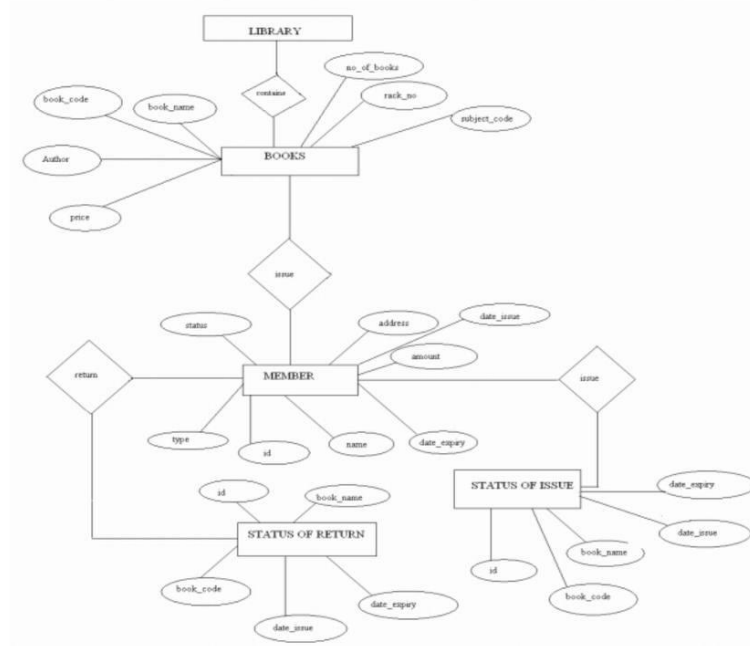
Question (6)

Marks (12)

A library is planning to develop system, Draw the Entity Relationship Diagram (ERD) for a library information system.

1. This library will serve only registered students.
2. Each book has title, author, publisher, date-of-publication, ISBN number.
3. Each student has SSN, first name, last name, address, and telephone number.
4. You have to specify the Entities, Attributes, Primary Keys, Relationships, Degree of the Relationship between different Entities.

Note: A student can borrow a book for up to two weeks, so it is necessary to keep track of the date-of-borrow. He can then renew the book for another two weeks. Two renewals are allowed. After two renews, the student must return a borrowed book. If a book is overdue, the student must pay a fine of S.R 0.5 a day.



Question (7) - Optional Marks(6)

Consider the following Entity Relationship (ER) diagram.

- Assume that an employee may work in up to two departments or may not be assigned to any department.
- Assume that each department must contain one and may contain up to three phone numbers.
- Assume that each department can have anywhere between 5 to 25 employees.
- Assume that each phone is used by one, and only one, department.
- Assume that each phone is assigned to at least one, and may be assigned to up to 10 employees.
- Assume that each employee is assigned at least one, but no more than 6 phones.

Write the (min, max) constraints for both sides of each relationship on the following ER diagram.

